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Childhood Cancer Survivors in Latin America: Insights into Health Outcomes and Information Needs

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Abbreviations key:

HICs	High-income countries
LMICs	Low- and middle-income countries
CCSs	Childhood cancer survivors
PAHO	Pan American Health Organization

CCSS	Childhood Cancer Survivor Study
CCSS-NAQ	Childhood Cancer Survivor Study Needs Assessment Questionnaire
REDCap	Research Electronic Data Capture
HSCT	Hematopoietic stem cell transplantation
CNS	Central nervous system

Abstract:

Background: Childhood cancer survivors (CCSs) face long-term health challenges, yet the health and specific needs of Latin American survivors remain underexplored. This study aimed to describe the health-related, psychosocial late effects, and information needs among CCSs in the region.

Methods: This mixed-method study combined quantitative data from an online survey with qualitative insights from follow-up interviews, involving CCSs from the largest regional survivor network and additional participants recruited through snowball sampling. We report the quantitative results here. The survey was adapted from the Childhood Cancer Survivor Study Survey Follow-Up and the Needs Assessment Questionnaire (CCSS-NAQ).

Results: A total of 135 CCSs from 16 Latin American countries participated (mean age 26.1 years). The most frequent late effects were mental health (61.2%), endocrine (58.6%), and cardiovascular (42.1%). Infertility was prevalent (16.9%), with nearly one-third unsure of their fertility status. Anxiety (51.1%) and depression (31.8%) were common. Significant findings included higher rates of endocrine disorders in females (p=0.0018) and those who had received radiation (p=0.048), respiratory problems in stem cell transplantation recipients (p=0.026), and auditory impairment in survivors of a solid tumor (p=0.0031). While most felt

informed about their health, only one-third had regular medical follow-up and access to psychological care. Survivors expressed strong interest in learning more about fertility and mental health, with a preference for digital information.

Conclusions: The study documents the need for region-specific survivorship care that addresses both the physical and mental health challenges of Latin American CCSs.

110

111 **INTRODUCTION**

112 Advances in risk-stratified and multimodal therapies have significantly improved
113 childhood cancer survival rates in high-income countries (HICs), rising from 60%
114 in the 1970s to over 85% today[1,2]. However, survival rates in low- and middle-
115 income countries (LMICs), where 80% of children with cancer live, remain lower
116 and more variable[3-5]. Most Latin American and Caribbean countries are upper-
117 or low-middle income, with Chile, Panama, and Uruguay as high-income
118 exceptions[6]. From 2000 to 2014, five-year survival rates for childhood cancers
119 varied widely across the region, e.g., acute lymphoblastic leukemia, which ranged
120 from below 60% in Ecuador to over 90% in Puerto Rico[7]. These disparities are
121 largely driven by limited access to treatment, lack of specialized care and delayed
122 diagnoses[8,9]. The absence of childhood cancer registries in many LMICs further
123 complicates efforts to track survival outcomes and long-term care planning,
124 though outcomes are improving with the inclusion of childhood cancer in national
125 cancer control plans[9-11].

126 Childhood cancer survivors (CCSs) face a heightened risk of long-term medical
127 and psychosocial effects[12]. Thirty years post diagnosis, nearly three-quarters
128 experience chronic health conditions, and almost half face serious health
129 issues[13-15]. These effects can profoundly impact survivors' quality of life[16-18].
130 Evidence from HICs underscores the need for comprehensive, risk-based follow-
131 up care, alongside raising awareness among survivors and healthcare
132 professionals about long-term health risks[19,20]. Unfortunately, optimal long-term
133 follow-up care can be costly. Integrating multidisciplinary care and adhering to
134 HICs-based guidelines can be challenging, even in well-resourced settings[21]. In

135 Latin America, limited survivorship programs and healthcare access disparities
136 hinder effective follow-up care[22]. Many survivors only seek medical care when
137 symptoms become noticeable, often facing barriers to timely treatment, as
138 primary healthcare providers are generally unfamiliar with surveillance
139 guidance[23,24].
140 Despite growing awareness, data on the late effects in Latin American CCSs
141 remain limited[25,26]. This study aimed to explore region-specific health and
142 psychosocial late effects, and information needs, to inform targeted public health
143 strategies aligned with the Global Initiative for Childhood Cancer, led by the World
144 Health Organization and the Pan American Health Organization (PAHO)[27,28].
145

146 **METHODS**

147 **Design and setting**

148 This mixed-method study integrated quantitative data from an online survey (in
149 Spanish and Portuguese) with qualitative insights from four open-ended survey
150 questions and a subset of follow-up interviews with survey respondents. Written
151 informed consent was obtained from all participants, and oral consent was
152 recorded before each interview. The study was approved by the Hospital Sant
153 Joan de Déu Research Ethics Committee (Protocol number: PIC-10-24,
154 approved January 25, 2024).
155

156 **Sample selection**

157 Participants were eligible if they had a history of childhood cancer diagnosed
158 before the age of 21, were aged 18 or older when completing the survey, had
159 been off therapy for at least five years, and were originally from or resided in a

160 Latin American country. Initial participants were identified through Faros de Vida
161 (Lighthouses for Life), the largest regional CCSs network, with additional
162 recruitment via snowball sampling. Due to the nature of the simultaneous
163 recruitment process, it was not possible to distinguish between participants
164 recruited through the CCSs network and those recruited through snowball
165 sampling.
166
167 **Data collection**
168 The survey, developed by experts in childhood cancer survivorship with
169 experience in Latin America (pediatric oncologists, nurses, psychologists, global
170 child health experts and CCSs), focused on survivors' physical and psychosocial
171 health, and information needs. The instrument was based on the Childhood
172 Cancer Survivor Study (CCSS) Survey Follow-Up and the Childhood Cancer
173 Survivor Study Needs Assessment Questionnaire (CCSS-NAQ)[29,30]. A total of 88
174 core questions were selected to align with the study's objectives. Some questions
175 were modified during translation into Spanish and Portuguese by native-speaking
176 healthcare professionals for clarity. The refined instrument consisted of four
177 sections: general information, current health status, psychosocial status, and
178 late-effects information.
179 A pilot test with five survivors from the regional network resulted in further survey
180 modifications for clarity and technical issues. The final version was distributed to
181 all CCSs in the Latin American network via a QR code sent through email and
182 shared via snowball sampling to additional survivors and Latin American pediatric
183 oncologists who were encouraged to forward it to former patients. The survey,
184 hosted on REDCap (Research Electronic Data Capture) at the Pediatric Cancer

Center Barcelona of the Hospital Sant Joan de Déu, was conducted from March to April 2024[31,32].

Statistical analysis

Statistical analysis was conducted using R statistical software (version 4.4.1)[33]. Descriptive statistics were employed to summarize demographic and health-related characteristics. Continuous variables were described using mean and standard deviations, as well as median, minimum and maximum, while categorical variables were described with absolute frequencies and percentages. Differences in health outcomes across groups were assessed using chi-square test or Fisher's test for categorical variables. Since no prior hypotheses were formulated, comparisons were exploratory. The level of statistical significance was set to 0.05.

RESULTS

Participant characteristics

A total of 293 survey responses were received, of which 135 (46.1%) met the inclusion criteria and were suitable for analysis (See Supplemental Fig. S1). Some participants did not answer every question, resulting in missing data. Total respondents for each item are shown in the tables. The study included participants from 16 Latin American countries, with the highest response rates from Brazil (22.4%, *n*=30), Ecuador (14.9%, *n*=20), and Mexico (14.2%, *n*=19) (Fig. 1). The mean age was 26.1 years (SD=5.9; range=18-48 years). Most participants (84.9%, *n*=112) were diagnosed with cancer before age 15, and 65.9% (*n*=89) identified as female. Table 1 summarizes participant demographic

210 and clinical data. The most common diagnoses were leukemia (43.6%, *n*=58)
211 and lymphoma (21.8%, *n*=29). Most respondents were single (81.3%, *n*=109),
212 just over half were university-educated (53.4%, *n*=70), and among the 91
213 participants not currently studying, 59.3% (*n*=54) reported stable employment.
214

215 **Health status**

216 Survey responses were categorized into 12 health-condition groups, with each
217 survey question assigned to a specific group based on medical criteria. To identify
218 the most prevalent late effects, a survivor was considered to have a health
219 problem in each category if they answered 'yes' to any symptom included in the
220 category. If all responses in a health-condition category were 'no', the survivor
221 was considered not to have a problem in that category. Respondents who
222 answered 'unsure' or who did not provide a response for a symptom were not
223 counted in that category, since their condition could not be determined. Table 2
224 summarizes the late effects by health-condition groups, and Supplemental Table
225 S1 provides the detailed late effects reported by participants. Responses were
226 additionally classified into three categories: 'No', 'Yes', and 'Unsure'. The
227 responses 'Yes, and the problem is still present' and 'Yes, but resolved' were
228 combined into 'Yes' category. This allowed us to identify individuals who had
229 experienced or were currently experiencing issues that required evaluation.
230 The most common late effects reported were mental health issues (61.2%, *n*=71),
231 endocrine disorders (58.6%, *n*=58), cardiovascular complications (42.1%, *n*=51),
232 and neurocognitive/academic challenges (31.9%, *n*=37). Five participants (3.8%)
233 reported a second cancer diagnosis, and 9.1% (*n*=11) experienced chronic pain.

234 Additionally, 52.5% (*n*=64) of survivors reported 'other' late effects, with dental
235 problems being the most frequent (42.9%, *n*=54).
236 Infertility (16.9%, *n*=22) and obesity (14.0%, *n*=18) were the most common
237 endocrine late effects reported overall. In female participants, amenorrhea
238 (25.9%, *n*=22) was the most frequently reported. One-third (31.5%, *n*=41) were
239 'unsure' whether infertility was a possible late effect. For cardiovascular issues,
240 tachycardia, or palpitations (17.7%, *n*=23) and dyslipidemia (16.6%, *n*=22) were
241 the most frequent, with 9.9% (n=13) reporting being 'unsure' about experiencing
242 chest pain or shortness of breath during exercise. In neurocognitive/academic
243 domains, memory problems (22.9%, *n*=30) and learning difficulties (16.0%, *n*=21)
244 were reported.
245
246 **Health status by demographics, treatment, and diagnosis**
247 Women had a significantly higher percentage of endocrine disorders compared
248 to men (69% versus 32.1%, *p*=0.0018). Respiratory problems were more
249 prevalent in individuals who had received hematopoietic stem cell transplantation
250 (HSCT) compared to those who had not received HSCT (57.1% versus 17.0%,
251 *p*=0.026), while endocrine disorders were more frequent in those who had
252 received radiation therapy compared to those who had not undergone radiation
253 therapy (71.1% versus 48.1%, *p*=0.048). Auditory impairment was associated
254 with surgical interventions compared to those who had not undergone surgery
255 (15.9% versus 4.2%, *p*=0.042) and was more common in survivors of solid
256 tumors than in those with hematological tumors (16.3% versus 3.7%, *p*=0.0031).
257 No other significant sex differences or associations with age at diagnosis were
258 observed. (See Supplemental Tables S2, S3 and S4)

Psychosocial status

In terms of psychosocial issues, 51.1% (*n*=67) of survivors reported anxiety, 31.8% (*n*=42) depression, and 17.8% (*n*=23) post-traumatic stress disorder. However, 10.7% (*n*=14) and 11.4% (*n*=15) were 'unsure' about experiencing anxiety or depression, respectively. A minority (14.6%, *n*=19) 'always' feared recurrence, 22.7% (*n*=30) 'always' experienced anxiety during health check-ups, and 33.3% (*n*=44) reported having anxiety 'always' or 'often' about their lives. Participants were asked about 'thoughts of death', and our intention was to assess for suicidal ideation or self-harm. However, since these specific terms were not explicitly included in the survey question, we cannot determine how respondents interpreted the question, or draw conclusions about the frequency of suicidal thoughts. Nevertheless, 10.7% (*n*=14) reported having 'thoughts of death' 'always' or 'often'. When combining responses from 'always', 'often', 'sometimes', and 'rarely', a total of 28.3% (*n*=37) reported experiencing such thoughts.

Regarding support needs, 9.8% (*n*=13) reported 'always' feeling the need for support to process cancer anxiety and fear, 4.5% (*n*=6) 'often', and 11.4% (*n*=15) 'sometimes'. Yet only one-third (35.7%, *n*=45) had access to psychological care; while 23% (*n*=29) reported if they needed care, they would not have access due to financial costs, scheduling, etc. Despite these challenges, 64.9% (*n*=85) reported being 'always' happy with life after cancer. Table 3 outlines the participants' psychological concerns. Despite the respondents' high education levels, 49.2% (*n*=61) reported academic difficulties they attributed to childhood

283 cancer treatment, and 59.7% (*n*=74) had not received educational support after
284 treatment.
285
286 **Information on late effects**
287 Seventy-eight percent (*n*=99) of CCSs reported that their physicians discussed
288 the need for long-term follow-up care due to potential late effects, either with them
289 or a family member. However, only one-third (33.3%, *n*=42) had a primary care
290 physician for regular visits, while 48.4% (*n*=61) reported seeing their physician
291 infrequently or only as needed. Approximately 62.5% (*n*=80) received a treatment
292 summary with future follow-up recommendations.
293 Respondents rated their level of agreement with statements about the health
294 information they received and interest in late effects. Responses were grouped
295 into 'agree'/'disagree', and 'interested'/'not interested' to identify general trends.
296 Most survivors agreed they were well-informed about their cancer history (93.5%,
297 *n*=116), while 69.6% (*n*=87) agreed they were aware of potential physical late
298 effects, and 65.6% (*n*=82) aware of emotional late effects. The majority (84%,
299 *n*=105) agreed they were informed about the necessary follow-up exams. Interest
300 in learning more about long-term health impacts was high; 88.7% (*n*=110)
301 expressed interest in physical and mental health effects, 95.1% (*n*=116) in quality
302 of life, 90.2% (*n*=111) in mental health, and 81% (*n*=98) in fertility. Social media
303 (61.5%, *n*=83) was the most preferred source for health information.
304
305 **DISCUSSION**
306 This study highlights the significant burden of late effects of childhood cancer
307 among Latin American CCSs, with nearly two-thirds reporting mental health

308 issues and endocrine disorders, and over a third experiencing cardiovascular
309 issues and neurocognitive or academic challenges. These findings are consistent
310 with research from HICs, emphasizing the enduring consequences of childhood
311 cancer on survivors' health and well-being[34-36].
312
313 Similar to findings from HICs, the mental health burden in this cohort is a
314 significant concern, with many participants reporting anxiety and depression[37-39].
315 Poor mental health outcomes among CCSs have been associated with several
316 risk factors, including low income, female sex, physical disabilities, and chronic
317 health conditions[40-43]. Many participants in this young cohort reported ongoing
318 medical issues that could potentially lead to chronic health conditions, suggesting
319 that these pre-existing factors may have contributed to the psychological distress
320 reported. Nevertheless, nearly two-thirds of survivors expressed overall
321 satisfaction with their lives after cancer, reflecting notable resilience. This contrast
322 can be understood through the psycho-oncological concept of post-traumatic
323 growth[44-45], which asserts that the extent to which survivors adapt is mediated by
324 the availability of resources that support the pursuit of new life goals, as
325 evidenced in the study conducted by Rosales et al. on Hispanic and non-Hispanic
326 white CCSs in California.[46]. However, individuals from lower socioeconomic
327 backgrounds or those with limited academic and occupational achievements may
328 face greater challenges in adjusting to life after cancer[46,47]. Given these
329 complexities, prioritizing mental health surveillance for this population is crucial[43],
330 recognizing CCSs as part of the broader group at-risk populations in Latin
331 America who require focused and sustained psychological support.

332 Our findings also reflect limited access to mental health services in Latin America.
333 Nearly half of the respondents expressed a desire for mental health support, yet
334 only a minority had access to a psychologist. This gap is consistent with
335 challenges in mental health resources across the region, including insufficient
336 funding and a lack of specialized professionals[48-50]. While these issues are
337 particularly pronounced in LMICs, studies from HICs have also identified barriers
338 to the utilization of psychosocial services, such as low income, limited insurance
339 coverage, and limited awareness of available resources[51,52]. In addition,
340 persistent cultural stigma surrounding mental illness–common across both HICs
341 and Latin America–may further discourage individuals from seeking mental
342 health support[53-56]. Interestingly, the presence of a primary care provider has
343 been associated with fewer barriers to accessing psychosocial care, suggesting
344 that integrating mental health services within primary care settings may offer a
345 feasible strategy to improve access in resource-limited contexts[51].
346
347 A high prevalence of academic challenges was observed in our cohort, as many
348 survivors reported difficulties in re-engaging with their studies post treatment and
349 identified a lack of adequate academic support. Although approximately half of
350 the survey respondents achieved university-level education, research has shown
351 that neurocognitive impairments and insufficient educational accommodations
352 can contribute to long-term professional difficulties with an impact on survivors'
353 economic stability[57-60]. A recent metanalysis showed that CCSs are significantly
354 less likely to advance through primary and secondary education or complete
355 tertiary education, especially those with central nervous system (CNS)
356 involvement[61]. In our study, only a minority of participants were CNS tumor

survivors, which may partially explain the relatively positive academic outcomes observed. Educational barriers often extend into the labor market, where CCSs face notable disadvantages[62]. Prior studies have shown that survivors may experience unemployment rates up to 50% higher than the general population[59,62,63]. In our cohort, the unemployment rate among respondents not currently engaged in academic studies exceeded the Latin American average of 6.4%[64]. Health-related limitations likely contribute to this disparity[65]; for example, a recent analysis of Nordic CCSs found that 9.2% of survivors were unemployed due to health-related reasons by age 30, rising to 14.3% by age 50[66]. These findings call for a collaborative effort among healthcare providers, and policymakers to develop supportive environments, including job training programs, screening for sensory and physical disabilities, provision of workplace accommodations, and access to disability benefits[67].

The health status of Latin American CCSs mirrors trends observed in HICs, with endocrine disorders being particularly prevalent[15,68]. In our cohort, infertility was among the most frequently reported endocrine sequelae and is widely recognized in the literature as a major quality-of-life concern for CCSs as they transition into adulthood[69-71]. Despite broad international consensus on the importance of fertility counseling and preservation at diagnosis, the implementation of oncofertility guidelines in Latin American remains limited[72,73]. Key barriers include financial constraints, limited healthcare coverage, and insufficient awareness among healthcare providers[72,74]. Cultural and religious beliefs may also influence attitudes toward fertility preservation, with some communities viewing reproductive interventions as morally or ethically controversial[74-76]. These factors

382 can lead to missed opportunities for timely intervention. In our cohort, a third of
383 respondents were unsure about their fertility status, reflecting the lack of fertility-
384 related information–a gap similarly documented in HICs[77,78]. Many survivors
385 receive limited education and guidance on reproductive risks throughout their
386 cancer trajectory, often becoming aware of potential gonadal toxicity upon
387 reaching reproductive age[70,79,80]. This uncertainty can contribute to significant
388 psychological distress[80,81]. While most participants in our study expressed a
389 strong interest in learning more about their fertility, these conversations can be
390 challenging for providers who are not trained in oncofertility care[82]. To be
391 effective, fertility discussions must be age-appropriate, culturally sensitive, and
392 tailored to the survivor's individual needs, ensuring that critical information is
393 delivered in an accessible and respectful manner[83,84].
394
395 Most participants in our study reported feeling well-informed about their medical
396 history and potential long-term effects, and many had received a treatment
397 summary with follow-up recommendations. Compared to data from HICs, our
398 results suggest relatively better access to information[85]. For example, a national
399 study from Norway found that at least 50-60% of survivors lacked written
400 information about their disease, treatment, and potential late effects[86], which has
401 been linked with missed follow-up care, higher healthcare costs, and increased
402 psychological distress[87-88]. Although our findings are encouraging within a LMIC
403 context, they should be interpreted with caution. Access to survivorship
404 information across Latin America is likely uneven, due to limited healthcare
405 infrastructure, and a shortage of healthcare professionals trained in the long-term
406 effects of childhood cancer[22]. The positive outcomes observed in our cohort may

407 reflect the efforts of the regional CCSs network actively promoting education on
408 late effects, rather than widespread healthcare support. Ensuring the consistent
409 delivery of survivorship information remains a critical challenge in the region.
410
411 Access to health care was available to only one-third of the respondents in our
412 study. Almost one-half reported infrequent follow-up visits, perhaps representing
413 a missed opportunity for early detection and management of late effects,
414 potentially exacerbating long-term health risks. Most participants expressed an
415 interest in learning about the enduring health impacts of cancer treatment, with a
416 noticeable shift towards digital health education. Research has shown that digital
417 access to treatment summaries and follow-up recommendations can empower
418 survivors to take an active role in their care[89]. Tools like the Survivorship Passport
419 (SurPass) and The Passport for Care have proven effective in supporting long-
420 term follow-up care in Europe and North America[89-92]. However, survivorship care
421 plans can also elicit emotional effects on CCSs, therefore they must be clearly
422 explained and integrated into long-term follow-up consultations[90,93].
423 Implementing such tools poses additional challenges in healthcare systems with
424 constrained capacities, as is often the case in many Latin American countries.
425
426 A major strength of this study is its region-specific focus, providing valuable
427 insights into the health status, access to long-term follow-up care, and information
428 needs of CCSs in Latin America, a population often underrepresented in
429 survivorship literature. The comprehensive nature of the study, which includes
430 both medical and psychosocial outcomes, allows for a holistic understanding of
431 the late effects experienced by CCSs in the region. Despite a relatively small

sample size that may not fully represent the broader CCSs population, the inclusion of participants from diverse geographic and socioeconomic backgrounds enhances the relevance of the findings across multiple healthcare systems in Latin America. However, this study also has limitations. The reliance on self-reported survey data mainly from a regional network may have introduced bias, as these participants are likely more engaged with follow-up care. Additional limitations include the absence of a control group and the lack of documented validation of the psychosocial measures included in the CCSS Follow-up Survey and the CCSS-NAQ in Latin American countries. Furthermore, the high proportion of university-educated respondents may have led to greater health literacy and better access to healthcare resources[94], potentially underestimating the true health burden, especially regarding unmet information needs. The cross-sectional design limits our ability to assess the long-term progression of late effects, and the relatively young mean age of 26 years may have hindered the capture of late effects with longer latency periods.

Conclusion

This study provides valuable insights into the long-term challenges faced by Latin American CCSs and contributes to the growing body of knowledge on survivorship care in the region, emphasizing the need for targeted interventions. Mental health challenges remain among the most prevalent and pressing issues for this population, with significant gaps in access to appropriate services and support. The disparities in oncofertility education for both healthcare providers and survivors underscore the need to establish this resource as a public health priority. Support includes not only providing fertility-related education to CCSs but

also ensuring the availability of affordable fertility preservation options for those at higher risk, both before and after cancer treatment. The study suggests that follow-up care could be strengthened by expanding the role of primary care physicians.

CONFLICT OF INTEREST

The authors declare no conflict of interest

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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LEGENDS:

FIGURE 1. Geographic distribution of the surveyed Latin American childhood cancer survivors. This map illustrates the regional participation and demographic spread of the surveyed childhood cancer survivors across Latin America.

TABLE 1. Demographic and clinical characteristics of the surveyed Latin American childhood cancer survivors

TABLE 2. Late effects by health-condition groups reported by the surveyed Latin American childhood cancer survivors

TABLE 3. Psychological concerns reported by the surveyed Latin American childhood cancer survivors

Supporting information material:

SUPPLEMENTARY TABLE S1. Detailed late effects reported by the surveyed Latin American childhood cancer survivors

SUPPLEMENTARY TABLE S2. Late effects by health-condition groups reported by the surveyed Latin American childhood cancer survivors by treatment received

SUPPLEMENTARY TABLE S3. Late effects by health-condition groups reported by the surveyed Latin American childhood cancer survivors by tumor type

SUPPLEMENTARY TABLE S4. Late effects by health-condition groups reported by the surveyed Latin American childhood cancer survivors by sex at birth

SUPPLEMENTARY FIGURE S1. Data inclusion flowchart. Participants were eligible if they had a history of childhood cancer diagnosed before the age of 21, were aged 18 or older when completing the survey, had been off therapy for at least five years, and were originally from or resided in a Latin American country.